

WEICHANG, China

WMO: 543110

Lat: 41.9605N

Lon: 117.7661E

Elev: 2932

StdP: 13.20

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -7.9	(c) -4.4	(d) -24.5	(e) 1.6	(f) 0.2	(g) -20.6	(h) 2.0	(i) 2.7	(j) 18.3	(k) 9.5	(l) 16.5	(m) 9.4	(n) 5.1	(o) 0	(p) 0.499

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.2	87.0	65.3	84.3	65.1	81.8	64.1	71.9	79.9	70.2	78.2	68.4	76.5	6.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.5	120.8	76.7	67.6	113.2	74.9	65.7	105.6	73.1	37.8	79.9	36.2	78.1	34.6	76.6	80.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.2	13.9	11.9	-12.8	92.7	4.8	3.3	-16.3	95.0	-19.1	97.0	-21.8	98.8	-25.4	101.2
			-13.5	74.2	4.6	2.4	-16.9	75.9	-19.6	77.3	-22.2	78.7	-25.5	80.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.7	10.7	17.5	30.7	46.0	57.5	65.9	71.3	68.2	58.2	44.2	27.3	13.6	
	DBStd	22.29	7.01	8.34	8.91	8.15	6.59	5.27	4.15	4.55	6.24	7.28	8.86	7.04	
	HDD50	4939	1219	909	599	171	15	0	0	0	11	205	681	1129	
	HDD65	8553	1684	1329	1063	572	245	51	4	21	215	644	1131	1594	
	CDD50	2282	0	0	1	50	247	478	659	565	256	26	0	0	
	CDD65	421	0	0	0	1	12	78	198	121	11	0	0	0	
	CDH74	3798	0	0	0	22	311	868	1526	917	151	3	0	0	
	CDH80	951	0	0	0	1	65	235	446	188	16	0	0	0	
Wind	WSAvg	4.8	5.1	5.0	5.5	5.7	5.6	4.4	4.0	3.9	4.2	4.4	4.5	5.0	
Precipitation	PrecAvg	16.6	0.1	0.2	0.3	0.6	1.4	2.6	5.1	3.8	1.8	0.7	0.2	0.0	
	PrecMax	25.5	0.4	0.4	1.2	2.0	3.2	4.4	13.0	8.8	3.5	2.4	0.6	0.4	
	PrecMin	9.8	0.0	0.0	0.0	0.0	0.3	0.7	0.7	1.1	0.2	0.0	0.0	0.0	
	PrecStd	3.7	0.1	0.1	0.3	0.4	0.8	0.9	2.7	2.1	0.9	0.5	0.1	0.1	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.3	45.8	64.2	77.2	86.8	90.2	92.5	88.3	82.4	71.3	54.6	37.4	
		MCWB	26.1	34.1	43.3	52.9	59.2	61.6	69.4	67.3	59.9	53.2	40.5	29.8	
	2%	DB	29.7	40.9	56.9	71.4	81.2	85.7	87.5	84.6	78.3	65.7	49.4	32.3	
		MCWB	22.7	30.2	40.3	49.5	56.6	62.3	67.1	65.9	60.1	50.2	38.5	25.2	
	5%	DB	26.4	37.0	51.4	67.4	77.3	82.5	84.6	81.8	74.8	61.7	45.2	28.8	
		MCWB	20.4	28.0	37.1	48.1	54.8	61.5	67.6	65.7	58.3	48.1	35.6	22.5	
	10%	DB	23.1	32.8	46.6	62.8	73.4	79.3	82.0	79.2	71.5	57.9	40.9	25.7	
		MCWB	18.0	25.2	34.0	45.7	52.6	60.6	66.8	65.0	57.2	46.1	32.7	20.5	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.8	35.1	45.7	57.2	63.1	68.9	74.7	73.6	66.9	56.7	42.6	29.7
			MCDB	33.4	42.2	61.8	70.5	78.7	80.2	84.1	81.3	76.5	65.8	50.7	37.6
2%		WB	22.9	30.5	41.1	52.6	59.8	66.6	72.7	71.4	63.5	53.2	39.3	25.5	
		MCDB	29.2	40.2	55.0	67.8	74.1	77.1	81.5	78.4	72.4	61.9	47.8	31.5	
5%		WB	20.4	28.1	37.8	49.5	57.6	65.0	71.1	69.5	61.3	50.5	36.3	23.0	
		MCDB	26.1	36.5	50.3	63.4	70.8	75.6	79.4	76.9	70.1	59.0	44.4	28.7	
10%		WB	18.0	25.4	34.8	46.7	55.4	63.5	69.6	67.6	59.2	47.6	33.3	20.5	
		MCDB	22.9	32.6	45.9	61.0	69.0	73.6	77.8	75.1	67.8	55.5	40.4	25.4	
Mean Daily Temperature Range	5% DB	MDBR	19.9	22.5	22.8	24.1	25.1	22.1	19.2	20.4	22.8	22.2	20.2	19.1	
		MCDBR	24.3	27.5	30.1	32.1	33.6	29.0	24.7	24.8	28.0	27.9	24.6	22.9	
		MCWBR	18.4	18.8	17.4	15.8	14.3	10.7	9.0	9.8	12.2	14.8	15.3	17.1	
	5% WB	MCDBR	24.0	26.5	28.1	27.0	26.3	20.7	18.0	18.1	21.5	23.6	23.5	22.6	
		MCWBR	18.5	18.4	16.7	14.3	12.2	9.1	8.3	8.9	11.1	14.0	15.2	17.1	
Clear-Sky Solar Irradiance	taub	0.294	0.361	0.407	0.477	0.538	0.610	0.573	0.530	0.481	0.425	0.331	0.302		
	taud	2.291	2.086	1.999	1.877	1.775	1.692	1.824	1.903	1.948	2.013	2.226	2.262		
	Ebn at Noon	268	260	262	252	239	222	229	235	236	233	248	252		
	Edh at Noon	28	40	50	61	69	75	65	59	52	43	30	26		
All-Sky Solar Radiation	RadAvg	806	1114	1502	1753	1949	1768	1710	1669	1449	1120	814	691		
	RadStd	46	77	96	122	132	156	109	99	84	75	64	38		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		-1.08	-2.48	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)		N/A	N/A	-1.96	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page